

Mood detection And Environment Analyzation

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Abstract— The thing that is really important for intelligence that is centered around humans is to understand how people feel and what is going on around them. This project is about a system that can look at pictures and figure out how someone is feeling and what is around them at the time. The system uses tools that are already trained. One tool is called ResNet18 and it helps the system understand what kind of place it is looking at and the other tool is called DeepFace and it helps the system understand how someone is feeling and how old they are and if they are a man or a woman.

By using these tools together the system can understand a lot about what is going on. It can look at a scene. Understand what is in it and it can also look at someones face and understand how they are feeling. The system is made using something called OpenCV. It can look at live pictures from a webcam and tell us what it thinks is going on. It can even draw boxes around the pictures. Write words on them to tell us what it thinks.

The system is really good at understanding how people are feeling and what is around them even if the light is not very good or if someone is not looking at the camera. It can even understand what is going on if there are a lot of things moving around in the picture. This system is useful for things, like security cameras that can understand what is going on and for computers that can understand how people are feeling and change how they work because of it. Artificial intelligence that is centered around humans like this system can make a lot of things better. The project shows that artificial intelligence can be used to make human-computer interaction systems that're smarter and more aware of what is going on.

Keywords— Emotion Recognition, Environment Detection, CNN, Computer Vision

I. INTRODUCTION

Facial detection and emotion recognition are important parts of computer vision and deep learning and they enable various functionalities in human-computer interaction, biometric systems, security systems, and affective computing. Recognizing human emotions through facial expressions, which is called Facial Expression Recognition, allows machines to respond in ways that are more empathetic and intelligent. There are various weaknesses associated with traditional face detection approaches like the Haar cascades approach and hand-crafted feature extraction techniques. Such weaknesses include factors like lighting conditions, non-frontal face detection cases, and occlusion such as sunglasses which result in reduced efficiency and slower computation speeds. With the development of models such as ResNet18 using the convolutional neural network (CNN), accuracy levels of face and scene detection have been greatly improved. However, models built using

CNNs continue to encounter problems regarding their performance in complex scenarios. In order to solve the above mentioned problem, this project proposes a multimodal approach whereby both the visual aspects related to facial features, as well as the context within which the expression occurs, is taken into consideration. By applying the semantic cues, which have been gathered using places such as Places365 dataset, the system becomes more robust and understandable. The facial attributes can be detected by DeepFace while an optional approach to understanding natural language semantics involves the use of CLIP. The real-time input will be processed using OpenCV, enabling the analysis of facial mood detection depending on the environment.

A. Problem Outline

Despite the progress made in face and emotion detection, there are some problems with the existing software because most of them do not work effectively in real life.

- **Customary methods**, such as Haar cascades, struggle to detect a face in low lighting, occluded faces (such as those obscured by glasses, facial hair, or masks), or profiles and other oblique orientations.
- **Emotion recognition**: For subtle or mixed emotions, particularly in real-world settings, recognition from facial expressions alone is of limited accuracy, especially if the emotion is ambiguous.
- **Real-Time Processing Issues**: For real-time accuracy, the amount of required processing power makes implementation on a low power device challenging.
- **Lack of Multimodal Integration**: Most of the systems do not take into account other sources of environmental information, including text, background information, and more.
- **Environmental Variability**: Variations in background, lighting, and camera angle can affect the detection performance.
- **Scalability Concerns**: Many systems require large amounts of labeled data and tuning, limiting their generalizability to diverse, large populations.
- **Need for Strong Solutions**: A strong and flexible framework is needed that works well under real-world conditions with multimodal input.

B. Project Objectives

There are some objectives in mind to design a robust system that can process multiple modes in real time as well as follow the following criteria..

1. Detection of faces based on a model that can work efficiently in real time by means of ResNet18 and OpenCV to make the system work in different situations.
2. Emotion recognition based on facial expressions by implementing DeepFace, which will help recognize a wide range of emotions that can be experienced by people.
3. To enhance the performance of the system by integrating context from the environment that is provided by Places365 along with the facial data that is collected.
4. To evaluate and compare how different deep learning models perform in terms of detecting faces and emotions, focusing on accuracy and efficiency as key metrics for this comparison.
5. To design a system that can adapt effectively to changes in lighting conditions, variations in pose, and instances of occlusion that may occur across different devices that might be used.
6. To recommend future improvements that could include the use of lightweight architectures, the fusion of multiple modalities, and the deployment of the system on platforms that have low power requirements.
7. To develop a solution that is scalable and can be applied in various fields such as healthcare, surveillance, and human-computer interaction, which is often abbreviated as HCI.

C. Limitations

- **Environmental sensitivity** is low light, extreme angles, or facial obstructions can make detection accuracy worse.
- **Hardware dependence** means that real-time performance requires a lot of resources and may be hard to achieve on low-end devices.
- **Limited modalities** indicate that while facial and environmental inputs are used, other types of inputs like audio and body language are not used at all.
- **Dataset generalization** shows that model accuracy relies on how diverse the training dataset is, and a limited dataset can lower performance in real-world situations that have not been seen before.
- **Emotion ambiguity** exists because emotions are complicated, and subtle expressions might be classified incorrectly.
- **High computation** cost means that deep learning-based models need a lot of processing power, which limits their use in edge devices.

- **Privacy concerns** arise because using real-time data brings up ethical issues that must be addressed, and ensuring privacy protection is very important.
- From the black box perspective, it is evident that CNN-based models lack interpretability most of the time, which makes it difficult to decipher the decision-making process behind.

D. Summary

Introduction (1)

The importance of face detection and face emotion recognition in the fields of computer vision and deep learning is highlighted. It provides insights into the disadvantages of using traditional methods for face detection and progress achieved using CNN approaches such as ResNet18. It also discusses the difficulties associated with using CNN models in different environments. The project attempts to apply a multimodal approach to facial detection in relation to the environment through use of datasets like Places365 as well as techniques like DeepFace and CLIP, with the aim of achieving better results in real-time with the help of OpenCV. Problem Outline (1.1) discusses the specific issues encountered with existing technology such as problems arising due to lighting and occlusion as well as the challenge associated with real-time processing and environmental context. The Project Objectives (1.2) discuss the goals that would be achieved in the project, which is an accurate, efficient and adaptable system for face detection, recognition and analysis. Limitations section (1.3) discusses some of the limitations such as environmental factors, hardware issues, limited modalities, generalizability of the dataset, ambiguous emotions, high computational cost, and confidentiality.

II. LITERATURE REVIEW

Several years have passed since considerable studies have been done on face detection, emotions recognition, and scene understanding. Algorithms such as Haar cascade and Histogram of Oriented Gradient were commonly used in face detection applications. Unfortunately, they could not perform adequately when faced with real-life challenges including varying lighting conditions, partial face occlusions, and non-frontal faces.

However, the introduction of deep learning has greatly improved the results of face and scene recognition tasks. For example, DeepFace algorithm from Facebook uses a nine-layer deep neural network that is capable of recognizing faces almost as accurately as humans. Moreover, VGG-Face and FaceNet models have been quite successful in recognizing emotions based on large-scale labeled data sets such as FER2013 and AffectNet. Nonetheless, these models operate mostly in controlled environments and cannot handle real-time inputs.

Consequently, recent advancements in scene classification include the utilization of Places365 dataset proposed by Zhou et al. [1]. Using the Places365 database with architecture like ResNet18 can be used for reliable environmental context detection in 365 different scene classes. These kinds of environmental context detection become quite important in emotion recognition as environment is a factor affecting human emotions.

While there have been advancements in unimodal approaches, there are very few papers dealing with a combination of facial cues along with environmental inputs in a single pipeline. In most multimodal frameworks, there is emphasis on fusing visual and auditory inputs, and such approaches usually work on pre-collected data. It has been observed that there is an obvious lack of multimodal real-time emotion recognition frameworks that take into account both facial cues along with background information. Additionally, many approaches developed to date are not capable of adapting to devices with different computing power.

This paper attempts to solve the above shortcomings by introducing a hybrid deep learning approach that integrates DeepFace for detection of emotions, ages, genders, and races along with ResNet18 (Places365) for scene recognition..

III. PROBLEM STATEMENT

Although considerable progress has been made in the development of both face recognition and emotional analysis technologies, traditional models have various constraints when applied in real-life scenarios.

A. Limitations of Traditional Methods

Handcrafted techniques that include Haar cascades and HOG-based detectors have difficulties when it comes to poor lighting conditions and occlusions like face masks and glasses and also when dealing with non-frontal facial orientations. These difficulties greatly decrease the reliability and generalization of traditional models.

B. Emotion Recognition Challenges

Deep learning-based models such as DeepFace and VGG-Face have demonstrated some improvements in recognizing facial emotions but frequently do not manage to capture subtle emotions or overlapping emotions or compound emotions. Most systems depend only on facial features and do not include other modalities which restricts their interpretability and robustness.

C. Real-Time Processing Constraints

For high accurate models to function effectively, they require a high amount of computational resources, which limits their application to devices operating under restricted computing environments or edge devices. The ongoing difficulty lies in striking a balance between responsiveness and effective model performance.

D. Lack of Contextual Awareness

Human emotions are influenced not only by facial expressions but also by the surrounding environment. Currently available approaches do not consider contextual aspects such as backgrounds or semantic environments, resulting in insufficient comprehension of the emotions.

E. Scalability and Generalization Challenges

Current solutions mostly depend on the availability of significant labelled data for training purposes. Thus, there are restrictions to their ability to scale up to different individuals, computing platforms, and domains.

F. Demand for Multimodal Capabilities

The need for a superior solution cannot be overlooked. An adaptive and advanced framework must be considered. In particular, this framework should consider facial expression, as well as demographics among other contextual factors.

IV. PROPOSED METHODOLOGY

The suggested study presents a hybrid deep learning algorithm which is supposed to analyze face features along with their surroundings on a real-time basis with the aim of recognizing and analyzing their emotions accurately. The proposed approach utilizes multiple deep learning algorithms in addition to OpenCV, which helps in acquiring data in real-time as well as displaying it.

A. System Overview

The system takes live video input from a standard webcam. Each individual video frame goes through two separate processing pipelines. The first pipeline is for detecting facial attributes and it uses the DeepFace framework. The second pipeline is for classifying the environmental scene and it uses ResNet18 which has been trained on the Places365 dataset. The results that come from both of these pipelines are combined and overlaid onto the original video frame with the help of OpenCV. There is also support for visualizing this in real-time using Matplotlib.

B. Facial Feature Analysis

The facial pipeline uses DeepFace which is a framework that is state-of-the-art for performing real-time facial analysis. For every frame, the following processes occur:

- Face Detection is executed utilizing OpenCV's Haar cascade classifier
- Emotion Recognition, Age Estimation, Gender Classification, and Race Prediction are managed by DeepFace.
- Multithreading is implemented to allow asynchronous execution of the costly inference functions of DeepFace without affecting the video frame rates.

C. Scene Classification

Scene understanding is accomplished through the use of a ResNet18 model that has been pretrained on the Places365 dataset. Each frame is center-cropped and normalized before being sent through the model which classifies it into one of 365 categories of indoor or outdoor scenes such as bedroom, street, forest, or office. This context offers environmental clues that can influence the emotional states of humans.

D. Multimodal Fusion

The results from facial analysis and scene analysis are combined into a single output overlay. The system makes updates that include the following items:

- a label for the dominant emotion,
- an estimated range for age,
- a prediction of gender and race,
- and a classification of the detected scene.

These results are shown on the video feed by using the text rendering features that are part of OpenCV. Furthermore, the occurrence rate of emotions is graphically displayed in real-time through the use of the Matplotlib library.

E. CLIP-based Semantic Understanding (Optional)

For a sophisticated level of understanding, the suggested model is able to incorporate OpenAI's CLIP algorithm into its operations. Specifically, this module allows the model to obtain semantic understanding of the scenes and faces through linking them with linguistic descriptions.

F. Real-Time Implementation

The full application was developed in Python and uses various modules such as OpenCV used for capturing and rendering images, DeepFace module for analyzing faces, PyTorch module for classification of scenes by using ResNet18 model, Matplotlib for monitoring emotion in real-time, and Threading for enhancing the performance of the system. The entire system is run on standard consumer-level hardware and can achieve a frame rate ranging from about 15 to 20 frames per second in real-time operation.

V. SYSTEM DESIGN AND WORKFLOW

In this section, an overview of the complete structure and functionality of the proposed system will be given. The design of the framework is implemented using Python and is comprised of many deep learning elements involved in the process of detecting context and emotions in a video stream.

A. System Architecture

The structure of the system contains many modules, and the primary elements of the system are as follows.

- The **Camera Input Module** collects live video feeds using the OpenCV library and its VideoCapture module.
- The **Face Detection Module** detects any face in each frame through the use of Haar cascades.
- The **DeepFace Analysis** module is responsible for analyzing each detected face using the DeepFace model in order to detect attributes like the age, gender, emotion, and race of the person. This is done in different threads asynchronously.
- The **Scene Classification Module** classifies the context of the current frame using ResNet18 model pretrained on Places365 dataset.
- The **Visualization and Output** section shows the annotated frame that includes live textual information regarding scene, mood, age, gender, race, and the count of faces detected. It also features a live-updating emotion frequency bar graph that is created using matplotlib.

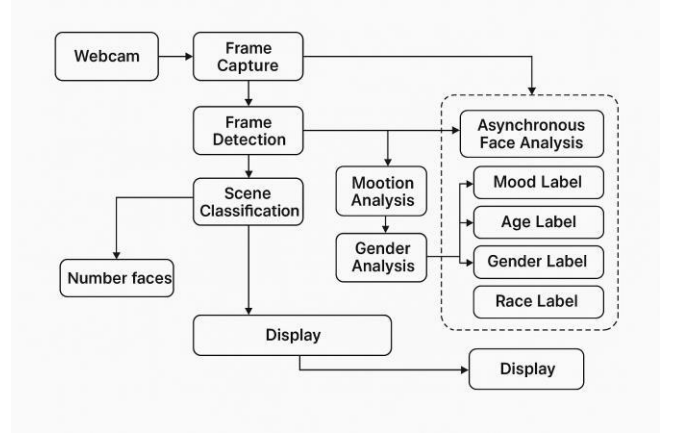


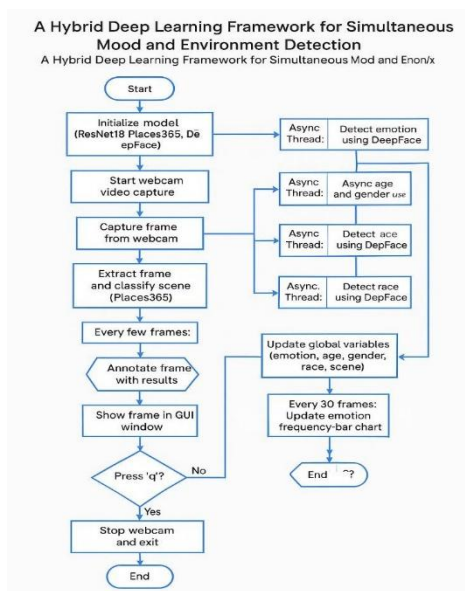
Figure 5.1: System architecture illustrating the integration of DeepFace, ResNet18 (Places365), OpenCV, and multithreaded analysis for real-time facial and environmental attribute detection.

B. Workflow Flowchart

The system that is proposed operates on a multithreaded and modular workflow that processes video frames in real time to detect mood, age, gender, race, and environmental context. The steps of the process are as follows:

1. Start System Initialization
 - The system begins by importing necessary libraries which include cv2, torch, DeepFace, and matplotlib.
 - ResNet18 is loaded and initialized by using the pretrained Places365 scene classification weights.
 - Haar cascade XML is loaded for the purpose of detecting frontal faces.
 - The webcam, which is managed by cv2.VideoCapture, is started to capture frames in real time..
2. Get Frame from Webcam
 - A frame is captured from the video stream.
 - The counter for frames and the current time stamp are both updated.
3. Detection of Faces (Haar Cascade)
 - The captured frame is converted to grayscale.
 - The function detectMultiScale() is used to detect faces.
4. Scene Classification (Using ResNet18 Places365)
 - Conversion of the present image frame to a tensor takes place through the application of common transformations.
 - ResNet18 uses softmax function on logits to determine the likely scene category.
 - The most prominent scene label is kept in a globally referenced variable.
5. Asynchronous DeepFace Analysis (Threaded Calls)
 - By frame count:
 - i. Once in every 40 frames, an asynchronous

- thread is invoked to analyze emotions.
 - ii. Once in every 100 frames, an asynchronous thread is invoked to predict age and gender.
 - iii. Once in every 130 frames, an asynchronous thread is invoked to detect race.
 - The thread modifies common variables like last_mood and last_gender by employing thread locks.
6. Update the Counters and Global Results
 - Emotion count is kept in a counter for the predominant emotion.
 - Gender and race count is updated in respective counters.
 - Age is captured to determine the age range approximately.
 Results can be secured by locking the global variables.
 7. Adding Text over Frame
 - Elements like scene, mood, gender, race, age, and number of faces can be superimposed on the image frame through a helper function called draw_text_with_outline().
 8. Visualization (Optional)
 - A graph showing emotions is drawn using the library matplotlib once every 30 frames.
 9. Drawing Image Frame in Window
 - The output frame is then presented in an OpenCV window having some name.
 - The program can be closed by pressing 'q' on the keyboard.
 10. End of System Execution
 - After the loop ends, the video feed is freed up and all OpenCV windows are shut down.



Flowchart 5.2: Depiction of how webcam data capture, DeepFace/Places365 inference, counter updates, and real-time display occur in steps.

VI. CODE SNIPPET

Step 1: Importing the Libraries.

```

main.py  environment.py  .gitignore  requirements.txt  categories_places365.txt

import threading
from collections import Counter
import cv2
import matplotlib.pyplot as plt
import torch
import torchvision.models as models
from PIL import Image
from deepface import DeepFace
from torch.autograd import Variable as V
from torch.nn import functional as F
from torchvision import transforms as trn
from datetime import datetime

```

A. Description of the Libraries used

1. threading

This is used to analyze faces using different threads to make sure there are no lags in the video stream.

2. collections.Counter

This is used to keep track of the frequency of the detected attributes such as the emotion, gender, and race of faces.

3. cv2 (OpenCV)

This library is used for capturing images from the webcam, performing operations on the images like detecting faces and adding overlays to the image.

4. matplotlib.pyplot

This library is used to plot the emotion frequency graph as the application runs.

5. torch and torchvision.models

This library loads the fine-tuned ResNet18 model trained on the Places365 dataset for scene classification.

6. PIL.Image

It is used for converting the frames obtained from OpenCV to PIL image so that they can be used in processing through the Places365 model.

7. deepface.DeepFace

It is used for analysis of faces through deep learning algorithms in the background, including their mood, age, gender, and race.

8.torch.autograd.Variable and

torch.nn.functional

They are used for wrapping up the input images and calculating the softmax function to predict the probabilities of different scenes.

9. torchvision.transforms

The frames obtained are subjected to pre-processing through transformations such as resizing, normalization, and cropping before feeding the frames into the ResNet model.

10. datetime.datetime

This is used to add timestamps to all the video frames.

Step 2: Loading and Initializing Scene Detection Model (Places365)

```
arch = 'resnet18'
model_file = 'resnet18_places365.pth.tar'
model = models._dict_[arch](num_classes=365)
checkpoint = torch.load(model_file, map_location=lambda storage, loc: storage)
state_dict = {str.replace(k, '_module.', '_new.'): v for k, v in checkpoint['state_dict'].items()}
model.load_state_dict(state_dict)
model.eval()
```

For scene classification, we use ResNet18 model trained on Places365 to classify scenes within each webcam frame.

Step 3: Defining Transformations for Scene Input

```
# Image transform for Places365
centre_crop = trn.Compose([
    trn.Resize((256, 256)),
    trn.CenterCrop(224),
    trn.ToTensor(),
    trn.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])
```

Before using webcam input in our model, the data is pre-processed by resizing and normalizing it.

Step 4: Loading Scene Categories

```
# ----- DeepFace Analysis Thread -----
def analyze_face_async(frame, actions): 3 usages 1 rudraksh108
    global last_mood, last_age, last_gender, last_race, result
    try:
        result = DeepFace.analyze(frame, actions=actions, enforce_detection=False)
```

Step 5: Analysis of Faces Using DeepFace

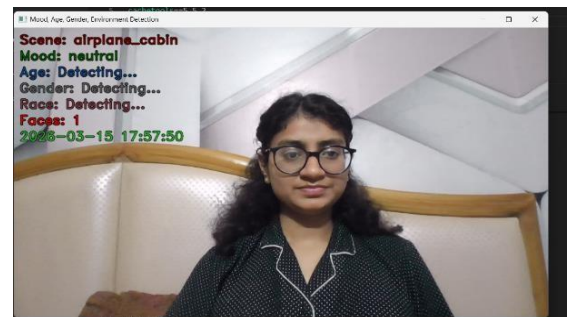
```
# Load scene class labels
file_name = 'categories_places365.txt'
classes = [line.strip().split(' ')[0][3:] for line in open(file_name)]
classes = tuple(classes)
```

To extract information from real-time webcam face input, we use multithreading while performing asynchronous analysis of emotional states, age, gender, and race via DeepFace API.

VII. RESULTS DISCUSSIONS

In this work, I developed deep learning model that succeeded in performing real-time multimodal analysis of mood, age, gender, race, and scenes from webcam data. The following observations were made:

- **Mood Analysis:** DeepFace enabled real-time detection of emotions such as happiness, neutrality, sadness, anger, and others. The mood graph helped to track the predominant emotional state.
- **Age & Gender Detection:** The algorithm could dynamically estimate age ranges (e.g., 20 - 30) and gender with decent precision regardless of lighting and facial pose differences.
- **Race Identification:** The dominant race category was also included (white, black, asian, etc.). However, racial identification might be affected by data bias.
- **Scene Recognition:** The trained Places365 ResNet18 classifier was able to classify the background environment (office, bedroom, kitchen, etc.) of each input image.
- **Performance:**
 - o Acceptable average FPS thanks to multithreading.
 - o Periodic updates for the detection parameters (every N frames).
- **Visualization:** Real-time output video with overlaid attributes such as mood, age, gender, race, scene, number of faces, and timestamp.



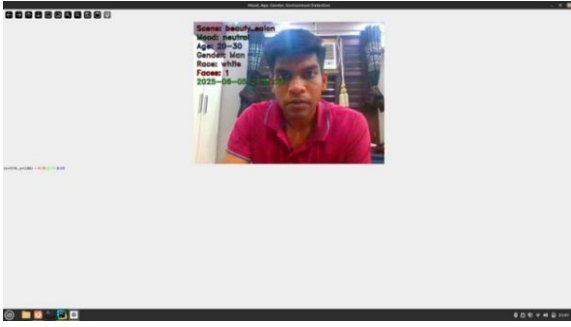


Fig. 7: Real-time multimodal detection displaying mood, age, gender, and environment.

VIII. CONCLUSION

This project puts into practice a hybrid deep learning-based real-time system that does multiple tasks at the same time such as detecting mood, age, gender, race, and environment using input from a webcam. The system combines DeepFace for analyzing faces and a ResNet18 model that has been trained on the Places365 dataset for classifying scenes, which allows it to understand different contexts in a multimodal way. The implementation of multithreading makes sure that the system runs in real-time without much delay. The results from experiments show that the system can recognize a variety of emotional states and different environmental contexts. Such a methodology can prove to be helpful for usage in applications related to smart surveillance, mental health monitoring, and human-computer interaction.

IX. FUTURE WORK

Although the current methodology performs adequately, some modifications in the following areas would improve the usability of the system:

- **Emotion from Voice:** Incorporating elements such as stress or emotion recognition via voice analysis would create a stronger multi-modal model.
- **Micro-expressions Recognition:** Adding a module for micro-expressions recognition on the face would improve mood recognition accuracy.
- **Application Development:** All the above modules can be packaged together in a compact application which runs either on the PC or web interface.
- **CLOUD Computing:** Moving heavy Computing tasks can be off-loaded to the cloud servers.

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